

Working with Meshes: **Surface Representations & Morphology Characterization**

- Volume rendering basics
- Marching Cubes – Surface renderings

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Basic processing steps for Rendering

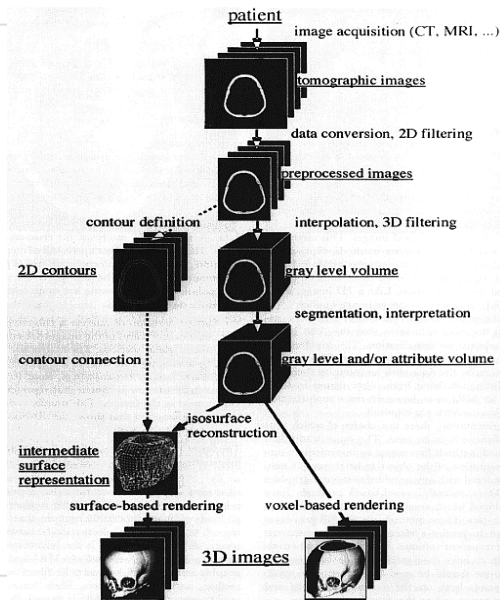
1. Data acquisition
2. Slices preprocessing and dataset preparation
 - good distribution of values, noise reduction, estimation of missing values, etc.
3. Data classification (usually iso-values)
4. Mapping the dataset into geometric or display primitives. Varies the most among algorithms

Two main techniques:

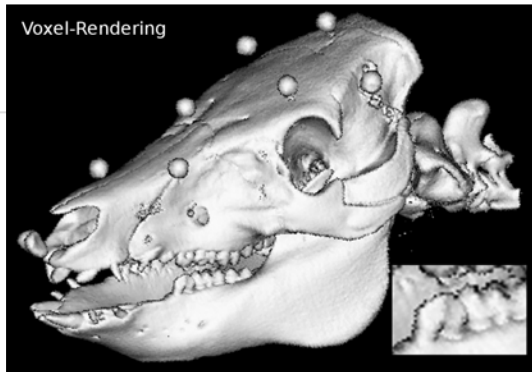
volume rendering and **surface rendering**

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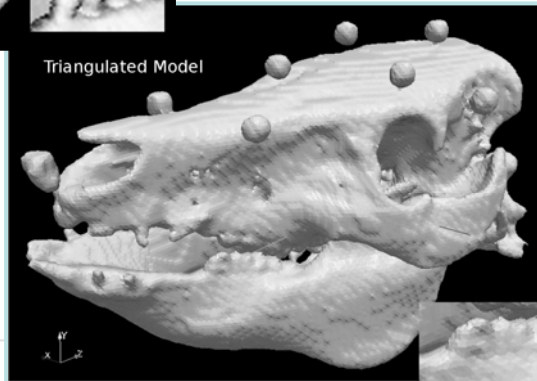
The imaging pipeline



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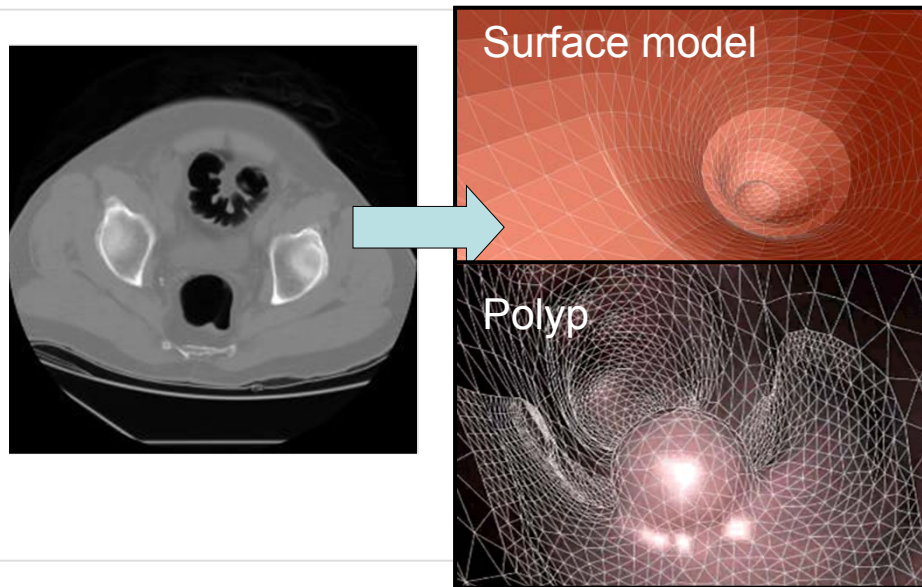


Voxel vs
triangulated
surface rendering



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Surface model from CT



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Volume visualization

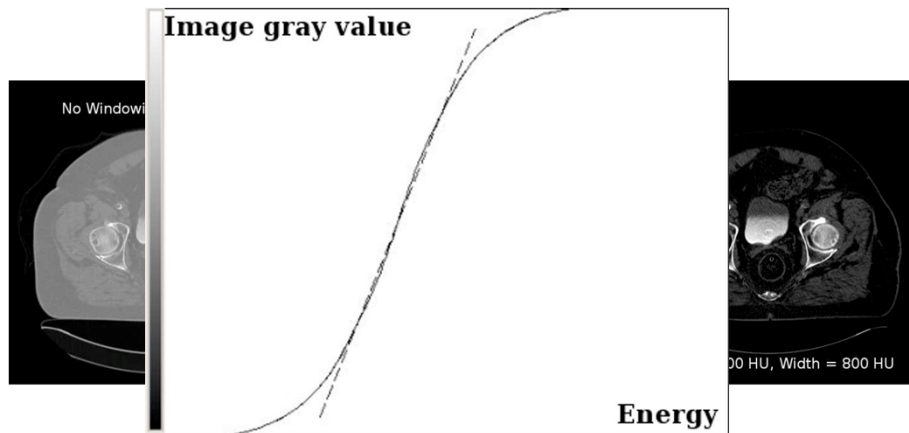
“The process of projecting a multidimensional dataset onto a two-dimensional image plane for the purpose of gaining an understanding of the structure contained within the volumetric data”

Must offer:

- understandable data representation
- quick data manipulation
- fast rendering

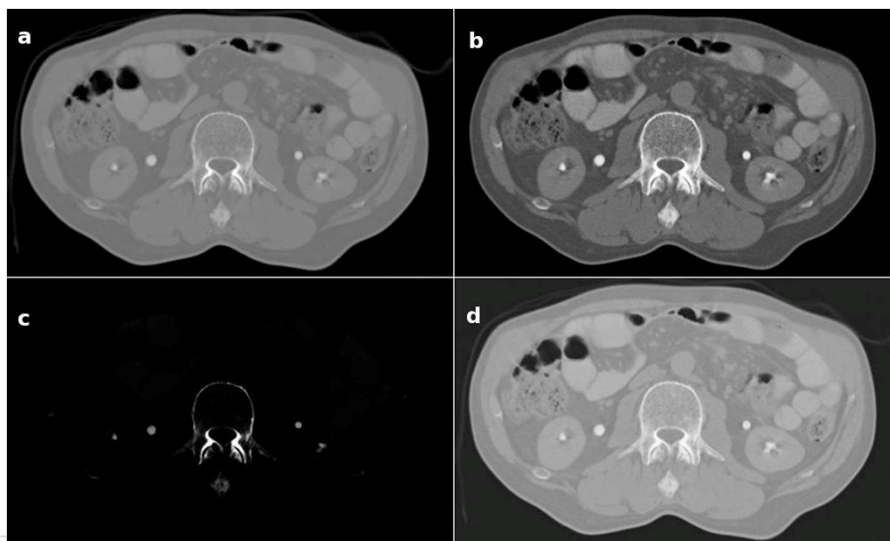
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2D visualization: windowing



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Windowing with sigmoid functions



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Biomedical Imaging (1330 / 2330)

Lecture 6, Fall 2016

Image Formation & Visualization

[*Imaging Systems Model Basics Text*: Ch. 1 and Ch. 2 (until 2.4, Linear Filtering) of *Insight into Images*

edited by Terry Yoo, et al. + *MRI Texts*: Joseph P. Hornak, *The Basics of MRI*

<http://www.cis.rit.edu/htbooks/mri/inside.htm> + Prince & Links, Ch 12 and 13 +

<https://www.youtube.com/playlist?list=PLD14D78BC61685BD7>]

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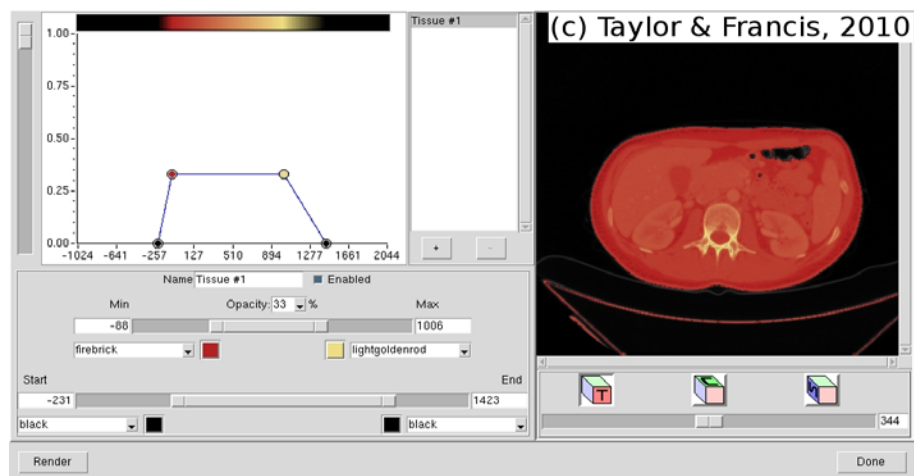
Lecture 6

September 15, 2016

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Opacity and color setting



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Ray-casting algorithm

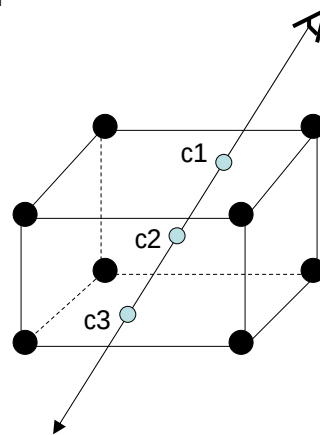
1. User sets up data-classification tables and viewing and lighting information for the scene
2. The algorithm runs a traversal of the image-plane pixels, assigning a color and opacity for each
3. A ray is fired from each pixel through the data volume
4. The opacities and shaded colors encountered along the ray are composited to give the opacity and color of the pixel

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Ray casting method

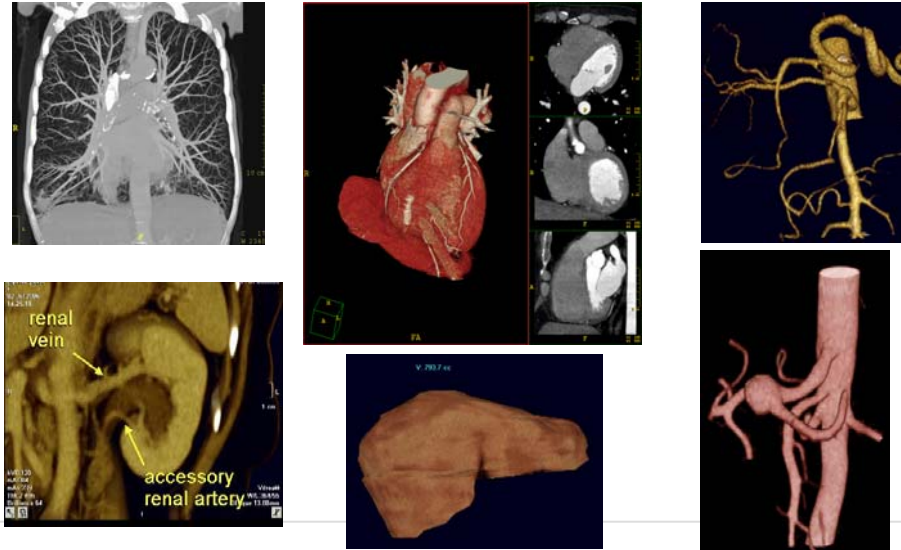
For each ray:

- Sample the dataset at evenly spaced locations along the ray.
- For each sample on the ray
 - Compute each sample value, e.g. by tri-linear interpolation.
 - Compute sample color and opacity by applying a transfer function.
 - Apply gradient shading on the sample color tuple.
- Compute the screen pixel color by compositing color tuples and opacities of all samples on the ray



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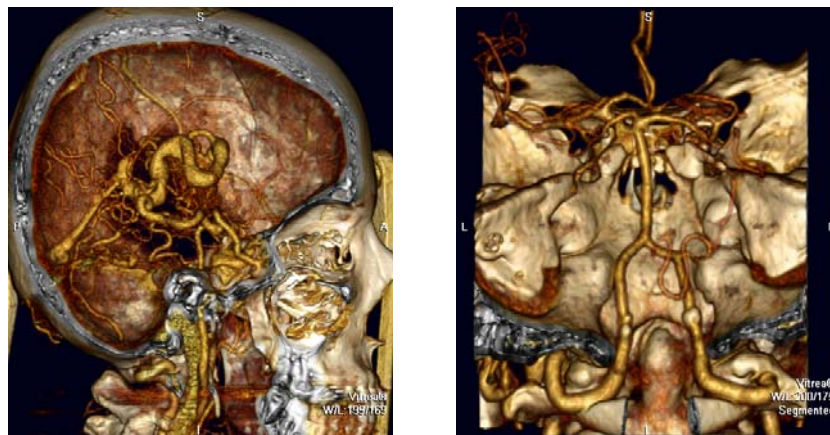
Volume-rendered images: Heart and vessels from CT Angiography.



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Volume-rendered images: Heart and vessels from CT Angiography.

Designed transfer functions for intensity set colors...



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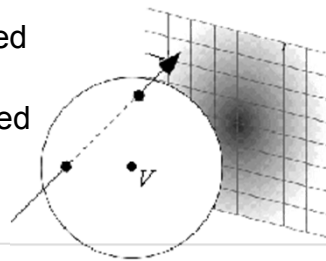
Splatting

Alternate method to Ray Casting.

- At each voxel a 3D spherical **reconstruction kernel** is defined (usually a 3D Gaussian)
- Projection of each kernel onto the image plane leaves a 2D **footprint** (splat) on the image plane
- Contribution of voxel V to image pixel P is

$$\text{footprint}_V(P) * (\text{color}_V, \text{opacity}_V)$$

- These contributions are composited into the image.
- The value at each voxel is classified according to user-specified color and opacity transfer functions before Splatting.

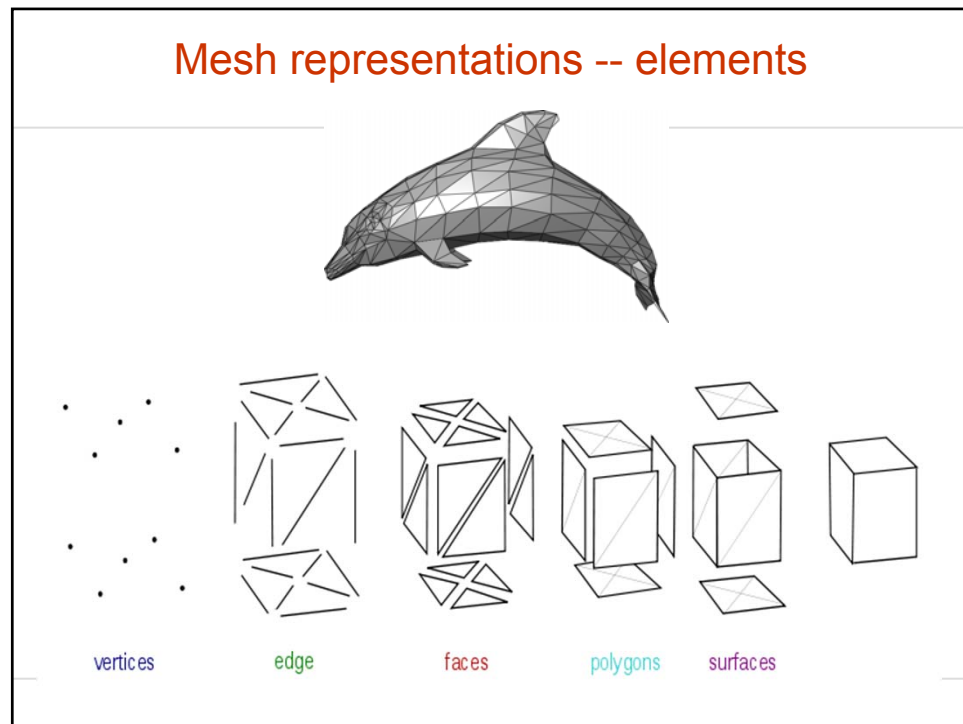


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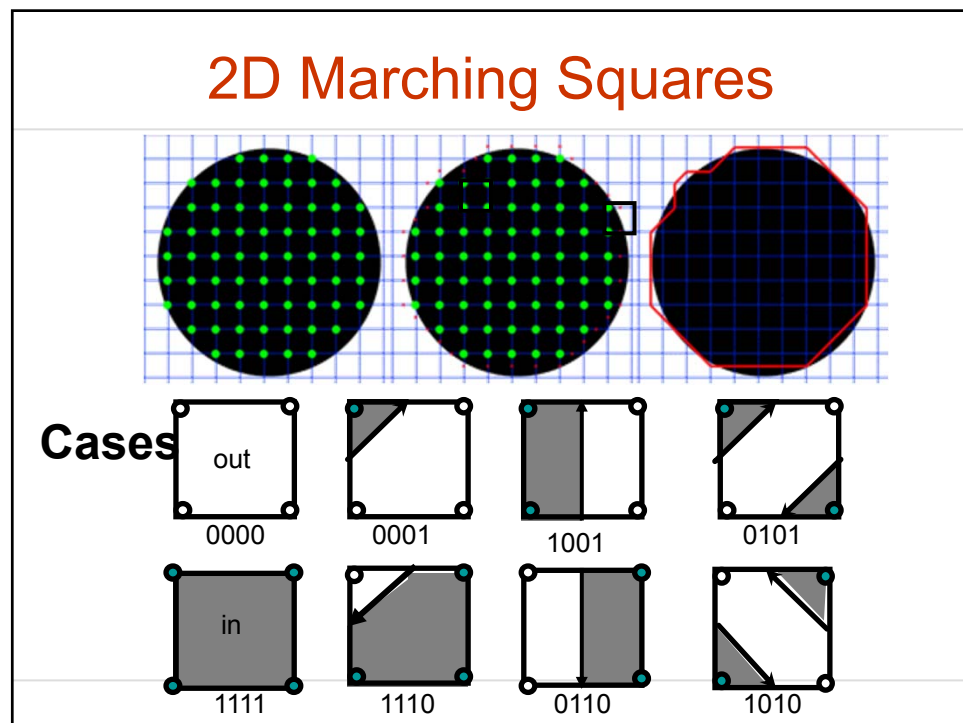
References

- Elvins, T.T. A survey of algorithms for volume visualization. *Computer Graphics*, 26(3):194-201, August 1992
- Levoy, M. Display of surfaces from volume data. *IEEE Computer Graphics and Applications*, 8(3):29-37, March 1988
- Westover, L. Footprint evaluation for volume rendering. *Computer Graphics*, 24(4):367-376, August 1990
- Smith, A.R. Image Compositing Fundamentals. *Microsoft Technical Memo 4*, August 1995
- Fattal, R., Lischinski, D. Variational classification for visualization of 3D ultrasound data. *IEEE Visualization 2001*.

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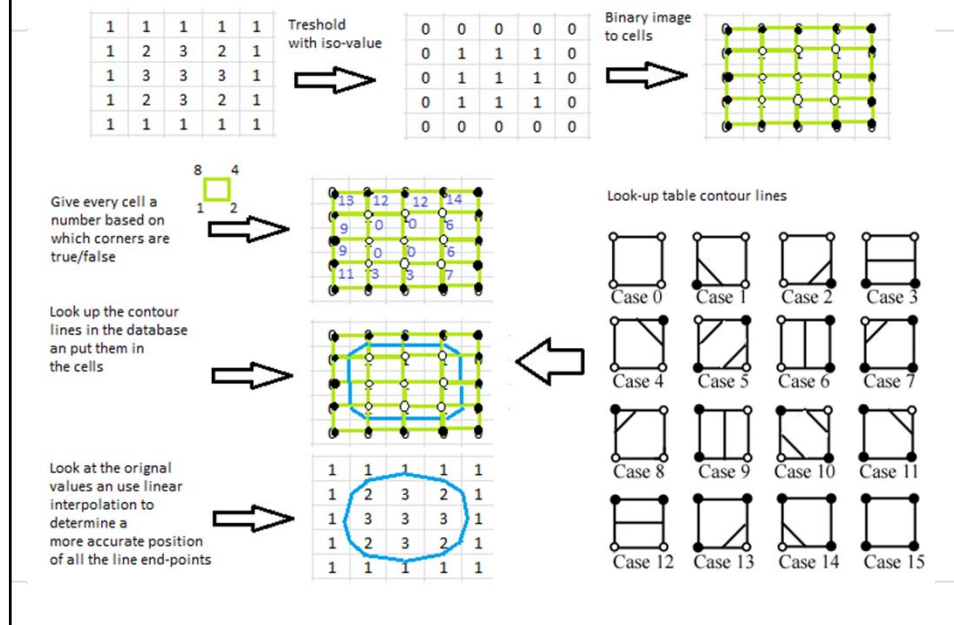


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Marching Squares: algorithm

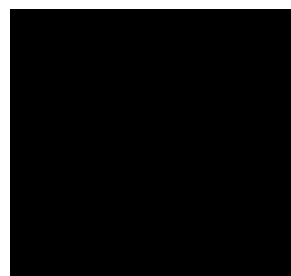


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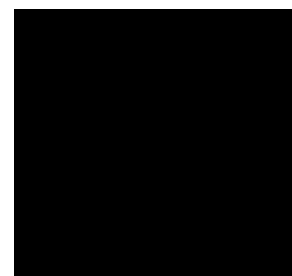
Marching squares: example

0	0	0	0	0
0	1	0	1	0
0	1	1	1	0
0	1	1	1	0
0	1	1	0	0

Segmented image



Graph nodes

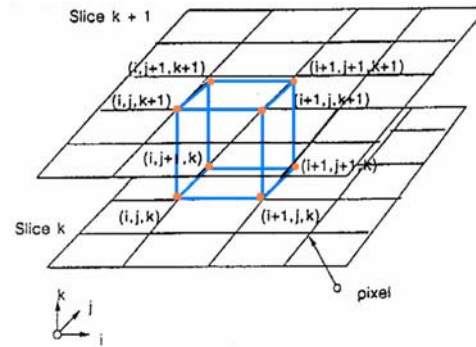


Reconstructed boundary

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3D: Marching Cubes Algorithm (Lorenson, Cline 1987)

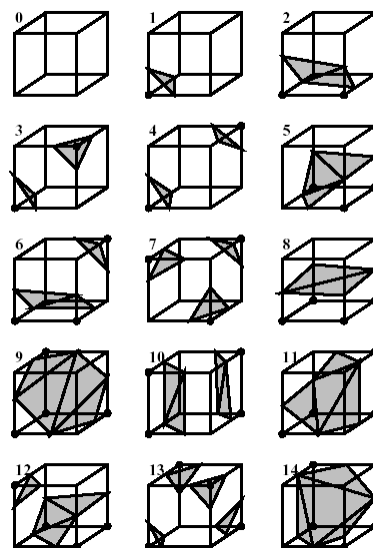
- Look at eight pixels in two consecutive slices
- Determine if a surface goes through according to the values at its edges
- If a surface goes through, construct the surface elements and "march on" to adjacent voxels.
- A cube with 8 vertices, each is either 1 (inside object) or 0 (outside)



Total of $2^8 = 256$ cases to examine (one bit to each vertex).

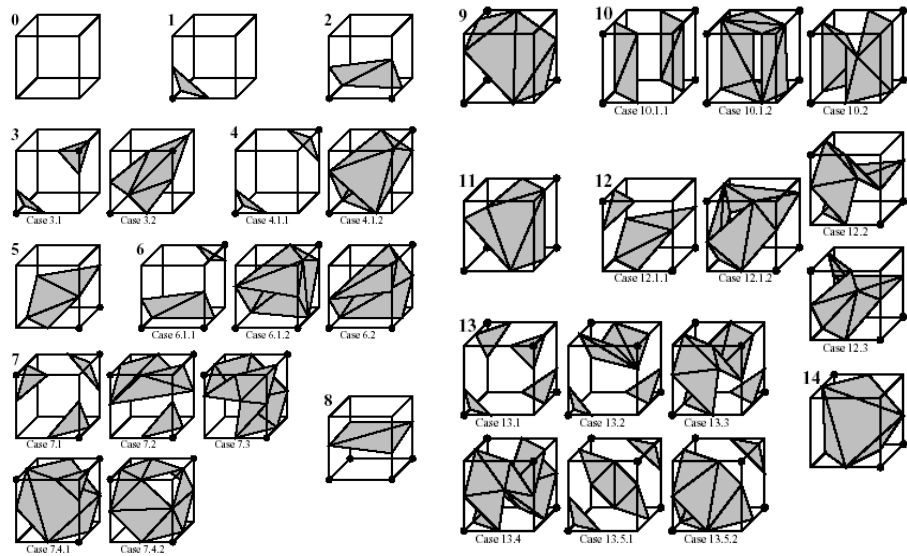
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Basic: 15 cases of surface topologies



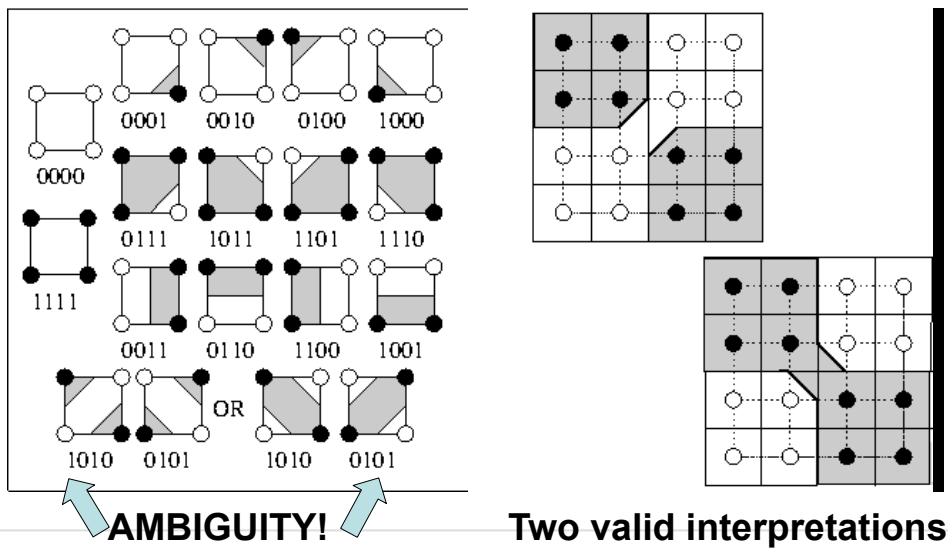
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Refined marching cubes: 33 cases



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Marching squares: ambiguity



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Surface normals

- Can help resolve ambiguities for Marching Squares / Cubes.
- Calculate unit normal for each triangle vertex (for shading calculations).
- Computed using the gradient vector – derivative of the image intensity function:

$$\bar{g}(x, y, z) = \nabla \bar{f}(x, y, z)$$